What Do LLM Judges Actually Verify? Matched Controls and Record Gaps for Document Currency

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Abstract

LLM judges are often asked whether the documents an answer cites are still in force, with an amendment record to consult. Such records are rarely complete. From a table of replacements, our test generates matched answers citing a replaced document, its successor, the replaced document before its replacement, and an older document that a newer one updated without replacing; it removes links from the record and scores each finding by its stated reason. On 48 replacements from the IETF RFC index, judges given a correct record caught nearly every replaced citation, as a lookup would. On the 24 faults whose link we removed, every judge caught fewer with the record than from the texts alone: Gemini 3.1 Pro 22/24 against 1/24, GPT-5.5 22/24 against 7/24, Claude Opus 4.5 18/24 against 10/24 and Gemma 4 26B 13/24 against 0/24. The drop recurred on a second random choice of removed links. Most misses carried no finding; rewording the gap as "no replacement recorded" did not restore the lost catches. Presenting the record as an unverified note, or warning that it was incomplete, restored most of them for Gemini, GPT-5.5 and Claude (significant after correction for the first two) but at most 5 of 13 for Gemma; where catches returned, so did false alarms on updates. With both texts, the judges mostly distinguished replacements from updates, and every update they flagged had a near-identical title. An appendix reports a descriptive case study on Indian rooftop-solar rules; we release the harness.

1 Introduction

Contracts are amended by later agreements, regulations by later notifications, and technical standards by later standards. An answer drawn from such a collection can quote a provision correctly and still be wrong, because the document it quotes was replaced before the date the question is about. Systems that use language models for this increasingly add a second model, an LLM judge, to check the answers before they reach a user. The judge is often given help: an amendment record listing which document replaced which, and from when.

Our starting point is a practical worry about that arrangement. Amendment records are seldom complete. Consolidated versions lag behind new notifications; pipelines that extract links from documents miss some; registers cover only the instruments they were built from. Wherever a record is silent about a replacement, it effectively reports a replaced document as current. If the judge reads the documents in front of it, such gaps cost little. If it relies on the record, it will pass exactly the answers the record cannot catch, and it adds little beyond what a lookup in the record would do.

Existing evaluations of LLM judges do not tell a builder which of these to expect. Benchmarks give the judge correct context or none (Tan et al. 2025; Xu et al. 2025), and most report whether the judge flagged an answer, not why. A judge that flags every older citation would score well on detection, and so would one that flags any older document that sits next to a newer one on the same subject. Neither is checking currency.

We therefore built a test aimed at one error, citing a replaced document, and at the question of what the judge relies on when the record and the documents disagree. The test starts from any table of replacements. For each replacement it writes an answer citing the replaced document (a fault) and three answers that are correct: one citing the successor, one citing the replaced document at a date before the replacement existed (a date control), and one citing an older document that a newer one updated

without replacing (a decoy). It then removes half the links from the record, presents the damaged record in several ways that leave its content unchanged, and counts a fault as caught only when the judge says the cited document was replaced.

We applied the test to 48 replacements recorded by the IETF's RFC Editor, with four judges (Gemini 3.1 Pro, GPT-5.5, Claude Opus 4.5 and the open-weight Gemma 4 26B). The hypotheses, tests and the way we would read each outcome were committed to the project repository after a smaller pilot on 24 of the links and before these runs.

With a correct record line, the judges caught every replaced citation or nearly so, which a lookup in the record also does. On the faults whose link we removed, every judge caught fewer with the record present than from the texts alone, on each of two random choices of which links to remove, and most of the missed answers were passed without any finding. The drop also occurred when the line said only that no replacement was recorded, a statement that is true. Presenting the record as an unverified note, or warning that it was incomplete, returned Gemini, GPT-5.5 and Claude most of the way to their text-only behaviour, false alarms included; the change is significant after correction for Gemini and GPT-5.5 but not for Claude, which had fewer catches to recover, and Gemma recovered at most 5 of the 13 it had lost (§5.2). Given both texts and no record, the judges caught many more replacements than they flagged updates, but every update they flagged had a near-identical title, and an unplanned analysis suggests that they treat the resemblance between two documents as evidence of replacement. Date controls, which the judges here nearly all passed, exposed in the pilot a 32B open model that flagged almost every old citation whatever its date.

The paper's contribution is the test, released as a small harness, together with this evidence about four judges on one corpus. We do not claim that LLM judges in general defer to records, and we cannot say why the judges we tested did. A second setting, Indian rooftop-solar regulation with an amendment record extracted by an AI pipeline, is reported as a descriptive case study in the appendix: its answer key did not pass our own quality checks, and nothing in the main results rests on it.

2 Related work

How LLM judges are evaluated.

Work on LLM judges has documented position, verbosity and self-preference biases (Zheng et al. 2023) and built benchmarks of harder judgements. JudgeBench (Tan et al. 2025) pairs responses that differ in correctness, and ContextualJudgeBench (Xu et al. 2025) asks judges to assess responses against supplied context, which it treats as the reference. Other studies probe what judges are sensitive to: they can be stable under edits that should not matter and insensitive to edits that should (Chen et al. 2026), and citation judges accept claims that their passage does not warrant (Qian et al. 2026). What counts as an error, or how it is scored, changes rankings: of citation verifiers (Kim et al. 2026) and of the models a benchmark evaluates (Pall 2026). Our test belongs to this line, with two differences in emphasis. The supplied context is a record that is deliberately incomplete, and findings are scored by the reason the judge gives, because in this task a flag for the wrong reason is easy to obtain.

Following a supplied signal over the evidence.

That models follow supplied signals is well established. Judges shown prior scores as metadata move toward them, and an instruction to disregard the metadata does not remove the effect (Kapetanovic et al. 2026). Code auditors given false assurances from static analysis lose detection, and prompt instructions do not reliably restore it (Liu and Gao 2026). Agents often adopt corrupted tool outputs (over a third of the time for every tool tested) and rarely warn users of the conflict (Yang et al. 2026). In the legal domain, models over-trust authoritative but false legal sources and struggle to tell whether a retrieved provision is still in force, and larger models appear more susceptible (Molfetta et al. 2026). The effect is not universal: judges given a wrong reference answer often ignore it (Lee et al. 2026), so how far a model follows a signal depends on the task and on the form of the signal.

Our results agree with this literature in direction. The setting differs in how the misleading signal arises. In the studies above it is constructed to be false: an injected score, a fabricated assurance, a corrupted tool output. Our record errs only through missing links, which is how real amendment records fail. In its authoritative form the gap then reads "in force", which is false, but in one condition the line says only that no replacement is recorded, which is true, and the judges' catches fell there too. The judge also has both documents in front of it, and a deterministic lookup in the same record gives a reference for what the record alone achieves. Our presentation conditions vary the record's wording, its apparent

authority and a warning about its completeness while keeping the record itself fixed; §6 compares the warning's effect with the anchoring result of Kapetanovic et al. (2026). Deference to an authoritative label is related to sycophancy towards a user's stated view (Sharma et al. 2024), but the source of the pressure is different.

Outdated evidence and temporal knowledge.

A separate line asks whether models that write answers can cope with change. Stale-Document Poisoning (Ahmed et al. 2026) shows that outdated retrieved documents overturn answers a model had right, and that an explicit statement of when the old evidence stops applying helps larger models. EvolveBench (Zhu et al. 2025) and earlier benchmarks of time-sensitive questions (Chen et al. 2021; Zhang and Choi 2021; Vu et al. 2024; Ouyang et al. 2025) measure whether models track facts that change, and work on knowledge conflicts asks when supplied context overrides what a model knows (Xie et al. 2024; Xu et al. 2024; Wu et al. 2024). These studies concern the model that writes the answer. We study a model that reviews an answer already written, which makes false alarms on correct answers part of the measurement.

Versioned law and amendment records.

Legal informatics has long modelled the change of norms explicitly: systems that manage consolidated acts across their successive versions (Palmirani and Brighi 2002), models of the temporal dimensions of norm change (Palmirani and Brighi 2006), defeasible-logic accounts of abrogation and annulment (Governatori and Rotolo 2010), and temporal metadata (in-force, efficacy and application times) in LegalRuleML (Palmirani et al. 2011). More recent work finds that retrieval over a corpus holding only current texts returns the wrong version of a provision (Cymbler et al. 2026), diagnoses staleness in statutory question answering (Prior et al. 2026), improves version-aware retrieval (Huwiler et al. 2025; Nam et al. 2026), and shows that LLM consolidation of statutes from amendment instructions is often not exact (50% exact matches for single amendments, 21% for chains) (Prior et al. 2025). Our record is a much thinner object than these models, a list of document-to-document replacements with dates, and our notion of currency ("replaced on or before the date asked about") is the simplest of the temporal notions they distinguish. We take such a record as given, imperfections included, and ask what a judge does with it.

Contrast items and right-for-the-wrong-reason.

Models can score well by exploiting features that correlate with the label (McCoy et al. 2019), and minimally different contrast items are a standard way to expose this (Gardner et al. 2020). Our date controls and decoys play that role for document currency. The date control changes only the date of a fault. The decoy keeps a newer document on the same topic, often with the same title, but one that leaves the older document in force. As §5.3 shows, our decoys control for title similarity but not for overlap between abstracts, so they are a partial contrast. Finally, label errors change model rankings (Northcutt et al. 2021; Gema et al. 2025), and a clinical benchmark built with LLM help was found to have at least 27% of its test labels likely wrong or incomputable (Ye et al. 2025). This is the main reason our primary data use the publisher's own record as gold, and the reason the regulatory study, whose AI-assisted key failed our checks, is reported only descriptively.

3 The test

The test needs two inputs: a record of replacements that is correct wherever it is used as gold, and a short text for each document. It needs no labelled answers, because the record already says which citations are wrong on which dates.

Matched items.

Every item poses the same kind of question, "As of date D, which document specifies T?", where T is the title of the older document, and supplies an answer that cites one document. For each link in the record (an old document and the document that replaced it) we write three items, and we pair each link with one decoy:

- a *fault* cites the old document, with D after the replacement;
- a *control* cites the replacing document, with the same D;
- a *date control* cites the old document, with D before the replacement existed, so the answer is correct;

Context	The judge is given, besides both texts
`both-texts`	nothing else
`status`	a line from the correct record: the cited document's status on the date
`status-damaged`	the line from a record with half its links dropped (a dropped link reads "in force")
`norecord-damaged`	as above, the line reading "no replacement recorded"
`lowauth-damaged`	as `status-damaged`, headed as an unverified note from an automated tagger
`caveat-damaged`	as `status-damaged`, with the warning that the record is incomplete and may omit replacements

Table 1 The six primary contexts. The four damaged contexts use the same damaged record; the warning adds one true statement about it.

- a *decoy* cites the older document of a different pair, in which a newer document *updates* the older one without replacing it. The older document is still in force, so this answer is also correct.

The fault and its date control are the same answer with a different date. A judge that objects to old citations without comparing dates will catch the fault and fail the date control. The decoy targets a different shortcut. In the fault, a newer document on the same subject sits next to the cited one; in the decoy, so does a newer document, often with the same title, but it did not replace the older one. A judge that flags any older document with a newer neighbour will catch the fault and fail the decoy. Catch rates alone cannot tell either judge apart from one that actually checks.

What the judge sees.

The context is the part a builder controls, and we vary it while holding the question and the answer fixed (Table 1). In every primary context the judge gets both documents' titles, dates and abstracts, with any sentence that states a replacement or names the other document removed, so that the replacement has to be inferred. Five of the six contexts add one line from an amendment record about the cited document. In the damaged record, half the links are dropped with a fixed seed; for a dropped link the record says nothing about a replacement, and its line therefore reports the replaced document as in force. We call the faults on these links *gap faults*. The four damaged contexts are built from the same damaged record and differ in how its line is presented:

- the authoritative form, an "amendment record lookup" reporting the document as `in_force`;
- the same record with the line worded `no_replacement_recorded` (wording);
- the same line headed as an "unverified note from an automated tagger" (authority);
- the authoritative form with a warning that the record is incomplete and may omit replacements (warning).

The pilot also used contexts that we did not repeat here: the citation alone, the cited document alone, masked document numbers, the replacing document's own notice of replacement, an instruction to check replacement first, and two worked examples. They are reported in Appendix A.

Scoring by reason.

The judge returns a verdict and a list of findings. Each finding names a document and gives one reason from a closed list: replaced, not yet published, not applicable, content error, unsupported, or other. We count a fault as *caught* only if some finding names the cited document with the reason that it was replaced. A judge that flags the fault as, say, a content error has noticed that something is off without identifying the error, and an any-finding count would credit it with the catch. On the three kinds of correct answer we record two false-alarm counts: a *validity* false alarm, a finding that a document was replaced or not yet published, and any finding at all. The first measures mistakes about currency; the second also counts complaints about relevance or content, which the question format may provoke (§5.4). A reply that cannot be parsed counts against the judge: as a miss on a fault and as a false alarm of both kinds on a correct answer. We call a reply on a fault a *bare pass* when it contains no finding at all.

Lookup baseline.

A deterministic lookup, with no language model, flags exactly the documents the record marks as replaced. With the correct record it catches every fault and raises no false alarm. With the damaged record it catches the faults whose link survived and none of the gap faults. It tells us what the record alone achieves, and so what a judge given the record adds.

Scope of the design.

The test covers one error, citing a whole replaced document, and one decision, the judge's verdict on a fixed answer. Partial amendments, chains of replacements and cross-references would need other item generators. Following the record can help as well as hurt. The gap faults measure the cost of following a silent record, and the decoys, whose record line is correct, give one measure of the benefit.

4 Data and setup

Replacements.

IETF Requests for Comments (RFCs) replace one another, and the RFC Editor's index (RFC Editor 2026) records each replacement ("Obsoletes" / "Obsoleted by") together with publication dates and abstracts. We kept the simplest cases: replacements in which the old RFC was replaced by exactly one RFC, the replacing RFC is itself current, was published in 2005 or later and replaces at most two RFCs, and both documents have abstracts. From these we drew 48 links with a fixed seed. Because the record is the publisher's, the gold needs no annotation. Each document is represented by its title, publication date and abstract. In 33 of the 48 links the two titles are near-identical (a normalised string-similarity ratio of at least 0.8), which matters for the shortcut analysis below.

Decoys.

The index also records updates ("Updates" / "Updated by"), in which a newer RFC amends part of an older one without replacing it. A decoy is an update pair in which the older RFC was never replaced and the newer one is current, chosen under the same filters as the links and sharing no document with them. For each link we took the nearest unused update pair within the link's same-title group, by a distance that combines IETF area (our proxy for topic), title similarity, both publication dates and both abstract lengths. The matching is greedy in link order and deterministic. Balance on the matched features is good for dates (standardised mean differences 0.04 for the old date, 0.02 for the new date and -0.02 for the gap) and acceptable for title similarity (-0.22); 37/48 decoys share their link's IETF area. Two imbalances remain. Updating RFCs have shorter abstracts than replacing ones (-0.55), a property of updates that we chose not to force away. More importantly, abstract overlap was not a matching feature. A replacing RFC usually repeats much of the abstract of the RFC it replaces (median similarity 0.80 for same-title links), whereas an updating RFC does not (median 0.17 for same-title decoys). We found this only after the runs; its consequences are discussed in §5.3.

Judges and runs.

Four judges are run on all six contexts with the same links and decoys: Gemini 3.1 Pro, GPT-5.5 and Claude Opus 4.5 through the Kaggle Benchmarks proxy, and Gemma 4 26B through Google's API at temperature 0. The proxy sets temperature and seed itself and does not report them, which limits exact reproduction of the hosted judges. New requests in these runs went through the proxy with an output cap of 4,000 tokens, sending the system and user text as two messages. Replies to prompts identical to those of the pilot were taken from the pilot's cache rather than requested again: all 72 shared `both-texts` replies of Gemma, GPT-5.5 and Gemini, and 70 of Claude's 72. For the hosted judges those replies had been obtained with the proxy's earlier call (one joined message, no cap). Reuse affects the `both-texts` items of the first 24 links, their `status` items (except for GPT-5.5, which had no pilot `status` block) and the damaged-record items whose prompt did not change. For Gemma, the client retries a request once without the API's JSON mode when the JSON-mode reply is cut off; the run log records that this happened, but not for which items. Every block of replies records the model version and a hash of the harness code. For this paper we rescored every reply from its raw text, and every stored score matched.

Hypotheses and analysis.

The hypotheses below, with their tests and the reading we would give each outcome, were written after the pilot and committed to the project repository shortly before the runs. This is a timestamped internal

record, not a public registration. Before any result existed, the plan was amended to add two presentation conditions (wording and authority, beside the warning) and to match decoys on more features; the amendment is part of the same history.

H1 Given both texts, a judge's right-reason catch on faults exceeds its validity false alarms on the matched decoys. Tested per judge as the mean paired difference over the 48 link–decoy pairs, with a 2,000-resample bootstrap interval over pairs.
H2a On the 24 gap faults, a judge catches fewer with the damaged record (authoritative form) than with no record. Two-sided exact McNemar test on the paired items; Holm adjustment across judges.
H2b Each other presentation of the damaged record changes the catch on the gap faults relative to the authoritative form. Two-sided exact McNemar tests; Holm adjustment across all 12 presentation–judge pairs.

The unit of analysis is the link, never the model call. With 24 pairs, a change in which every discordant item moves the same way needs at least six items to reach the 5% level before adjustment (two-sided exact $p = .031$), and more after the Holm adjustment; changes that do not reach it are reported as inconclusive. The plan also listed Wilson intervals for rates and an exploratory mixed-effects model pooled over judges; we report exact counts and paired tests instead, and did not fit the model, since no claim depends on it.

A second draw of removed links.

Because the pilot had shown that the choice of removed links matters, we amended the plan, once seed-0 results existed for three judges and before any seed-1 call, to repeat H2a on a second random choice (damage seed 1) for the same four judges, with the tests fixed in advance: H2a per judge on seed 1 with Holm adjustment across judges, and a per-judge estimate over both seeds. The two draws drop 24 links each and share 12, so each adds 12 gap faults of its own. A shared link is the same item under both seeds, with an identical prompt and therefore the same reply, so the pooled estimate counts it once: it is the mean drop in catch over the 36 links in either gap set, with a 2,000-resample bootstrap interval over links and an exact sign test on the links that changed. Text-only prompts do not depend on the record, so seed 1 reuses the text-only replies; its damaged-record replies on the 12 new gap faults were new requests (eight per hosted judge) except where the pilot had asked the identical prompt. The second draw is a robustness check on the choice of removed links, not an independent replication: it shares half its gap faults and all text-only replies with the first. The hypotheses and thresholds were not changed. Everything else in §5, including every comparison by title or abstract overlap, is exploratory.

The pilot.

Before the final design, six judges ran up to ten contexts on the first 24 links (Appendix A). The pilot had no decoys and no presentation conditions, and its damaged record, built over 24 links rather than 48, dropped a different set of links. We use it for the date-control result and for the contexts not repeated here.

The case study.

The test was first developed on Indian rooftop-solar regulation, with an amendment record extracted by an AI pipeline and seven judges. Because its answer key failed the quality checks set for it, it is reported descriptively in Appendix D and the supplementary material, and no result in §5 uses it.

Use of AI assistants.

An AI coding assistant, built on models from the same developer as one of the judges (Claude Opus 4.5), drafted the hypotheses and analysis plan, which the authors reviewed and amended; wrote the harness; sampled the links and decoys; ran the Gemma judge and prepared the hosted-judge runs, which were executed through the benchmark proxy under the lead author's account; analysed the replies; and drafted and edited the manuscript. In the case study it drafted the labels and extracted the amendment record, which two authors checked blind with a third adjudicating. The RFC gold is the publisher's, and the assistant's own model was not run as a judge. The authors checked judge replies, statistical results and references against their sources, read and revised the manuscript, and take full responsibility for its content.

Table 2 All 48 links, per judge and context. Catch: right-reason catches of faults (any: any finding on the fault). False alarms (FA) count validity findings (replaced or not yet published) unless marked any. Controls cite the replacing RFC; date controls cite the replaced RFC as of a date before its replacement; decoys cite an RFC that a newer one updates but does not replace. All counts are of 48.

		Faults		Controls		Date controls	Decoys	
Judge	Context	Catch	any	FA	any	FA	FA	any
Gemma 4 26B	`both-texts`	31/48	31/48	0/48	19/48	0/48	6/48	6/48
	`status`	47/48	47/48	0/48	20/48	0/48	0/48	0/48
	`status-damaged`	23/48	23/48	0/48	20/48	0/48	0/48	0/48
	`norecord-damaged`	23/48	23/48	0/48	20/48	0/48	0/48	0/48
	`lowauth-damaged`	26/48	26/48	0/48	18/48	0/48	1/48	1/48
	`caveat-damaged`	29/48	29/48	0/48	18/48	0/48	1/48	1/48
GPT-5.5	`both-texts`	45/48	46/48	1/48	12/48	0/48	8/48	8/48
	`status`	48/48	48/48	0/48	13/48	0/48	0/48	0/48
	`status-damaged`	31/48	31/48	0/48	13/48	0/48	0/48	0/48
	`norecord-damaged`	34/48	34/48	0/48	10/48	0/48	0/48	0/48
	`lowauth-damaged`	48/48	48/48	1/48	11/48	0/48	10/48	10/48
	`caveat-damaged`	47/48	48/48	1/48	9/48	0/48	8/48	8/48
Gemini 3.1 Pro	`both-texts`	42/48	42/48	0/48	21/48	0/48	6/48	6/48
	`status`	48/48	48/48	0/48	20/48	0/48	0/48	0/48
	`status-damaged`	25/48	25/48	0/48	20/48	0/48	0/48	0/48
	`norecord-damaged`	28/48	29/48	0/48	19/48	0/48	0/48	0/48
	`lowauth-damaged`	42/48	42/48	0/48	19/48	0/48	4/48	4/48
	`caveat-damaged`	46/48	46/48	0/48	20/48	0/48	6/48	6/48
Claude Opus 4.5	`both-texts`	38/48	39/48	1/48	19/48	2/48	9/48	9/48
	`status`	48/48	48/48	0/48	21/48	0/48	4/48	4/48
	`status-damaged`	34/48	36/48	0/48	21/48	0/48	4/48	4/48
	`norecord-damaged`	35/48	36/48	0/48	20/48	0/48	5/48	5/48
	`lowauth-damaged`	41/48	42/48	0/48	20/48	1/48	9/48	9/48
	`caveat-damaged`	41/48	42/48	0/48	20/48	1/48	7/48	7/48

5 Results

Table 2 lists every judge's catches and false alarms in every context. We begin with the correct record, which sets the baseline, and then turn to the gap faults.

5.1 With a correct record

Given a line from the correct record, the judges caught all or nearly all faults (Gemma 47/48, GPT-5.5 48/48, Gemini 48/48, Claude 48/48), as a lookup in the same record does by construction. The record also helped on correct answers. For decoys, the record line correctly reports the older RFC as in force, and decoy false alarms fell from 8/48 to 0/48 for GPT-5.5, from 6/48 to 0/48 for Gemini, from 6/48 to 0/48 for Gemma and from 9/48 to 4/48 for Claude. Following the record therefore has a benefit as well as the cost measured below, and on this corpus a judge given a correct record does essentially what the record does.

With the damaged record the lookup catches the 24 faults whose link survived and none of the 24 gap faults. Over all 48 faults the judges scored Gemma 23/48, GPT-5.5 31/48, Gemini 25/48 and Claude 34/48. The gap faults themselves are analysed next.

5.2 Gap faults

Table 3 isolates the 24 gap faults; the items are the same in every column, so each comparison is within a judge. One qualification applies to all four judges. Their text-only replies to the items of the first 24 links, which include 11 of the gap faults, are the pilot's replies, reused because the prompts are identical (§4); for these items the comparison is a partial replication of the pilot, not an independent one. On the other 13 gap faults, where every reply is new, the drop is of similar size: Gemini 12 to 1, GPT-5.5 12 to 5, Claude 11 to 6, Gemma 8 to 0.

From the texts alone, Gemini caught 22/24 of the gap faults and, with the damaged record in its authoritative form, 1/24: 21 catches lost and 0 gained (H2a, Holm p <.001). GPT-5.5 went from 22/24 to 7/24 (15 lost, 0 gained; Holm p <.001) and Claude from 18/24 to 10/24 (8 lost, 0 gained; Holm p .008). Gemma caught fewer from the texts, 13/24, and none with the record (Holm p <.001). H2a therefore holds for all four judges. Of Gemini's 23 misses under the damaged record, 23 were replies with

Table 3 The 24 gap faults per judge: faults whose link the damaged record dropped (half the 48 links, seed 0). Same items in every column. The four damaged-record presentations use the same damaged record. Bare: passes with no finding under the authoritative "in force" line. *p*: exact McNemar, Holm-adjusted; H2a compares the authoritative line with no record, H2b each other presentation with the authoritative line.

	No record	Damaged record, presented as					Holm *p*			
Judge	(texts)	record, "in force"	"no replacement recorded"	unverified note	record + warning	Bare	H2a	wording	authority	warning
Gemma 4 26B	13/24	0/24	0/24	2/24	5/24	24	<.001	1.000	1.000	.375
GPT-5.5	22/24	7/24	10/24	24/24	23/24	17	<.001	1.000	<.001	<.001
Gemini 3.1 Pro	22/24	1/24	4/24	18/24	22/24	23	<.001	1.000	<.001	<.001
Claude Opus 4.5	18/24	10/24	11/24	17/24	17/24	12	.008	1.000	.125	.125

Table 4 H2a on two random draws of dropped links. Each seed drops 24 of the 48 links; the two draws share some links (Shared). Per seed: gap faults caught with no record → with the damaged record (authoritative form), lost/gained, Holm-adjusted exact McNemar *p* (across judges, within seed). Pooled: unique links in either gap set, a shared link counted once (its reply is the same under both seeds); mean drop in points with a 95% bootstrap interval over links, lost/gained, exact sign test. Seed 0 is the planned test; seed 1 and the pooled estimate were added before any seed-1 run, when seed-0 results existed for three judges.

	Seed 0			Seed 1				Pooled over unique links			
Judge	Caught	Lost/gained	*p*	Caught	Lost/gained	*p*	Shared	Links	Drop	Lost/gained	*p*
Gemma 4 26B	13/24 → 0/24	13/0	<.001	17/24 → 0/24	17/0	<.001	12	36	+61 [+44, +78]	22/0	<.001
GPT-5.5	22/24 → 7/24	15/0	<.001	24/24 → 8/24	16/0	<.001	12	36	+64 [+47, +78]	23/0	<.001
Gemini 3.1 Pro	22/24 → 1/24	21/0	<.001	21/24 → 3/24	18/0	<.001	12	36	+81 [+67, +92]	29/0	<.001
Claude Opus 4.5	18/24 → 10/24	8/0	.008	20/24 → 13/24	7/0	.016	12	36	+31 [+17, +44]	11/0	<.001

no finding at all; GPT-5.5's were 17 of 17, Claude's 12 of 14 and Gemma's 24 of 24. Claude's two misses that did carry a finding both mention the newer RFC, but as a content error (a possible data error in the evidence, or an answer that may be incomplete), not as a replacement; one repeats, as the record says, that the older RFC remains in force. No reply names a conflict between the record and the texts.

A second draw of removed links.

Table 4 repeats H2a on the second draw. Each judge lost catches again and gained none: Gemini 21/24 to 3/24, GPT-5.5 24/24 to 8/24, Claude 20/24 to 13/24 and Gemma 17/24 to 0/24 (Holm p <.001, <.001, .016 and <.001). The two draws share 12 gap faults, whose replies are identical under both, so the seed-1 result is new evidence only through its other 12 items. Counting each link once, the drop over the 36 links in either gap set is +81 points for Gemini (95% interval [+67, +92]), +64 for GPT-5.5 ([+47, +78]), +61 for Gemma ([+44, +78]) and +31 for Claude ([+17, +44]), with no link gained by any judge. On both draws Gemini lost the most catches and Claude the fewest, but with this many items we do not rank the judges.

Presentation.

Writing "no replacement recorded" instead of "in force" did not bring any judge back to its text-only level. Gemini caught 4/24 (3 gained, 0 lost), GPT-5.5 10/24 (3 gained, 0 lost), Claude 11/24 (3 gained, 2 lost) and Gemma 0/24; no change is significant before or after adjustment. Small gains are within what 24 pairs cannot resolve, so the result is that the rewording did not restore the lost catches, not that it had no effect. This condition matters for one more reason. In it the record line is literally true: it reports only that no replacement is recorded. The drop seen here, for judges that had the replacement in front of them, is therefore produced by an omission in the record, without any false statement. (The comparison with the texts alone was not a planned test; for each judge every item that changed moved away from a catch.)

The other two presentations moved three of the judges. With the line headed as an unverified note, Gemini caught 18/24 gap faults and GPT-5.5 24/24; with a warning that the record omits some replacements, 22/24 and 23/24. Every item that changed moved toward a catch (Gemini 17 and 21 items, GPT-5.5 17 and 16), and all four tests are significant after the Holm adjustment over the twelve presentation tests (p <.001 each). Claude moved the same way, to 17/24 and 17/24 (7 items each, none lost; raw p .016, Holm p .125). In each condition it caught six of the 8 faults it had lost and one it had missed from the texts (Appendix B), almost its text-only level of 18/24; the test is not significant because Claude had few catches to recover, not because the change was small. Gemma recovered less, catching 2/24 and 5/24 against 13/24 from the texts alone (Holm p 1.000 and .375).

Where the catches came back, they came back with the judge's text-only false alarms. Under the authoritative record, Gemini and GPT-5.5 raised no validity false alarm on decoys, because the record correctly lists the decoy's older RFC as in force. With the unverified note, GPT-5.5 flagged 10/48 decoys and Gemini 4/48; with the warning, 8/48 and 6/48, against 8/48 and 6/48 from the texts alone. Claude's

Table 5 Shortcut test with both texts and no record. Faults and their matched decoys, split by whether the two RFCs of the pair share a (near-)identical title. Discrimination: catch minus decoy false alarm, paired by link, in points with a 95% bootstrap interval over the 48 pairs.

	Same title		Different title		
Judge	Fault catch	Decoy FA	Fault catch	Decoy FA	Discrimination
Gemma 4 26B	29/33	6/33	2/15	0/15	+52 [+35, +69]
GPT-5.5	32/33	8/33	13/15	0/15	+77 [+65, +88]
Gemini 3.1 Pro	31/33	6/33	11/15	0/15	+75 [+60, +88]
Claude Opus 4.5	30/33	9/33	8/15	0/15	+60 [+44, +75]

rose from 4/48 to 9/48 and 7/48 (texts alone 9/48). A decoy's older RFC was never replaced, so in every presentation its record line is correct; a judge that weighed record against texts would have no reason to flag more decoys when told the record might omit replacements. The change therefore looks more like a return to text-only behaviour than like better use of the record. False alarms on controls and date controls stayed at two of 48 or fewer in every condition.

Comparison with the pilot.

In the pilot, Claude kept 10 of the 11 gap faults it had caught from the texts. Here it lost 8. The two runs used different gap faults: the pilot's damaged record was built over 24 links, and only 5 of its 12 gap faults are gap faults here. On those 5 the replies are the same reused replies and Claude caught 3; the other 7 pilot items, all of which it caught, are not gap faults in this run. Most of the difference therefore comes from which links were dropped, and on the second draw Claude again lost catches (7) and gained none. Because the shared replies were reused rather than requested again, this comparison cannot speak to run-to-run variation, which the pilot's one repeated block suggests is real at the level of individual replies. A dozen items can therefore mislead, and comparisons between judges on item sets of this size, including those in our own tables, are provisional. Appendix B lists every gap fault.

5.3 Replacements, updates and surface similarity

Table 5 reports the decoy test, with both texts and no record. Every judge caught more faults than it flagged decoys (paired difference +77 points for GPT-5.5, 95% interval [+65, +88]; Gemini +75, [+60, +88]; Claude +60, [+44, +75]; Gemma +52, [+35, +69]), so H1 holds. H1 is a weak test, though: a judge that flagged every older document with a near-identical title would pass it as long as most replacements share their predecessor's title. The more informative comparison is within the same-title pairs, where title similarity is held roughly constant. There GPT-5.5 caught 32/33 faults and flagged 8/33 decoys, Gemini 31/33 and 6/33, Claude 30/33 and 9/33, Gemma 29/33 and 6/33. Among different-title pairs, no judge flagged a decoy, and catches differed widely (GPT-5.5 13/15, Gemini 11/15, Claude 8/15, Gemma 2/15).

Our plan read a positive H1 as evidence that text-only catches are not a title shortcut. The data call for a more qualified reading, and the analysis that supports it was not planned. Title similarity travels with abstract overlap: in the text the judges saw, a replacing RFC usually repeats much of its predecessor's abstract (median similarity 0.80 for same-title links, 0.15 for different-title links), while the decoys, which were not matched on overlap, do not (0.17 and 0.10). Splitting at a similarity of 0.3, chosen after seeing the data, false alarms on same-title decoys were more frequent when the abstracts overlapped (GPT-5.5 3/7 against 5/26; Gemini 2/7 against 4/26; Claude 3/7 against 6/26; Gemma 2/7 against 4/26), and so were catches on different-title replacements (Claude 5/6 against 3/9; Gemma 2/6 against 0/9; GPT-5.5 6/6 against 7/9; Gemini 6/6 against 5/9). The high-overlap cells hold six or seven items, and we report no test. The pattern is consistent with judges using the resemblance between the two documents as evidence of replacement. In the RFC series that is often good evidence, so we hesitate to call it a shortcut, and the data cannot separate the title from the abstract.

The decoys also have a labelling caveat. Some of the decoys the judges flagged are updates that introduce a new version of a protocol, such as RFC 8210 (version 1 of the RPKI-to-Router protocol, updating RFC 6810), flagged by all four judges, and RFC 9777 (MLDv2, updating RFC 2710), flagged by two. By the publisher's record the older RFC is in force, which is our gold. For a question asking which document specifies the protocol, however, a reader could reasonably prefer the newer version. Some of these false alarms may therefore be defensible judgements rather than errors.

5.4 Scoring by reason

For the frontier judges given both texts, counting any finding on a fault instead of the right reason barely changes the catch (GPT-5.5 45/48 against 46/48, Gemini 42/48 against 42/48, Claude 38/48 against 39/48). On correct answers the difference is large. With both texts, Claude raised some finding on 19/48 of the controls that cite the replacing RFC, but a validity finding on 1/48; Gemini raised some finding on 21/48 and a validity finding on 0/48; Gemma raised some finding on 19/48 and a validity finding on 0/48. Gemma's findings on these controls all gave the reason "content error". We suspect, without having tested it, that the question format provokes them: the question names the old document's title, and the answer cites a document with a different number and sometimes a different title. Under any-finding scoring these three judges would appear to raise false alarms on about two in five of these controls, almost none of which concern currency.

Without evidence the difference also affects catches. In the pilot, from the citation alone, Gemini raised a finding on every fault but gave the right reason on 11/24. Reason coding is also what makes the bare passes of §5.2 visible as a separate category from passes with an irrelevant finding.

5.5 Date controls

The judges raised few or no validity false alarms on date controls (with both texts: GPT-5.5 0/48, Gemini 0/48, Claude 2/48 and Gemma 0/48). In this run the controls did not separate the judges. In the pilot they did. Qwen 2.5 32B caught 42/48 faults with both texts, a catch rate comparable to the frontier judges, but also flagged 43/48 answers that were correct on their date, for a currency discrimination of −2 points against +92 for Gemini. Two worked examples, one of them a correct answer passed, reduced its date false alarms to 5/48. The result rests on one judge, and a catch rate alone would not have revealed it.

6 Discussion

The gap result.

Within each of the four judges, the same gap faults were caught less often when a record line was present, and most of the lost catches became passes with no finding. Some mundane explanations can be set aside. The items are identical across conditions. The drop also occurs when the line says only that no replacement is recorded, so it does not depend on the record stating anything false. And the decoy test shows that most text-only catches were not reflexive objections to older documents. What the design cannot set aside, the small and partly shared item sets and the reused replies, is discussed in §8.

We cannot say why the judges behaved this way, and at least three explanations fit what we observed. The judge may simply be relying on the record as the task presents it. It is shown an "amendment record lookup", nothing suggests the record is incomplete, and trusting it is defensible; on correct answers, following it removed most decoy false alarms (§5.1). On that reading the failure belongs to a record that does not disclose its own gaps, and what the judge lacks is any sign of registering the conflict with the texts. The line may instead be more salient than the evidence, being short, explicit and placed just before the task, while the replacement has to be inferred from two abstracts; we did not vary position or length. Or the judge may treat the status line as part of the question rather than as evidence to weigh. A bare pass is compatible with all three, and it does not tell us whether the judge noticed the conflict and resolved it in the record's favour or never noticed it.

Presentation.

The presentation conditions were meant to separate these explanations, and they partly do. Rewording the gap did not restore the catches for any judge, which lends no support to the simplest account, that judges are misled by an explicit claim of currency, although 24 pairs cannot exclude a small effect of wording. For Gemini and GPT-5.5, lowering the record's standing or warning that it was incomplete restored most or all of the lost catches, and Claude recovered six of the eight it had lost, without reaching significance. This fits the first explanation: told that the record might be wrong, these judges went back to the texts. It fits the salience explanation less well, since the note and the warning leave the line as short, explicit and close to the task as before. For these three judges, however, the decoy false alarms came back with the catches, to about their text-only level. The judges did not come to weigh record against texts item by item; they moved from relying on one source to relying on the other. Gemma moved least, catching at most five of the 24 gap faults however the damaged record was presented. Which

judges respond in this way, and why, is a question for more judges and more wordings than our one variant per condition.

Two related findings point in different directions. Kapetanovic et al. (2026) found that a warning did not reduce the total effect of judges' anchoring on prior scores, though it improved accuracy in one of their experiments. Our warning differs in stating a specific and true defect of the record, with the contradicting evidence in view, and either difference could matter. Molfetta et al. (2026) report that larger models defer more to authoritative but false legal citations, mainly among larger reasoning models. Our one open-weight judge responded least to the presentation of the record, but the hosted judges' sizes are not published, and four judges could not test an effect of size in any case.

Resemblance between documents.

The decoy results contain two observations that should be kept apart. The judges did not flag older documents merely for having a newer neighbour, since they caught far more replacements than they flagged updates. Yet their mistakes in both directions tracked how alike the two documents looked. Given both texts, updates were flagged only when the titles were near-identical, and Gemma rarely caught a replacement whose title had changed. Some of this reliance is reasonable, since RFC replacements often do repeat their predecessor's title and abstract, and some flagged decoys are new protocol versions that a reader might prefer to cite. The practical concern is transfer. An amended regulation may keep its title, while a contract amendment can look nothing like the agreement it modifies. A judge evaluated on one kind of collection may behave differently on the other, and nothing in a single-corpus evaluation, ours included, would show it.

Scoring.

Reason codes matter most on correct answers: under any-finding scoring, Claude, Gemini and Gemma would have appeared to object to roughly two in five controls, nearly all for reasons unrelated to currency. They also make the gap result measurable. A judge that calls a stale citation a "content error" has at least objected, and only reason-coded scoring separates it from a judge that says nothing.

7 Practical implications

These suggestions come from four judges on one corpus. Each is cheap to check with one's own judge, since the harness runs a judge through the six contexts in about 1,200 calls.

An evaluation in which the judge receives a correct record says little about the judge: with a correct record our judges did what a lookup does. The more useful test removes links from the record and compares each judge with itself on the affected items. If its catches there fall to the lookup's level, the judge contributes nothing beyond the record on those items, and the completeness of the record becomes the real limit of the pipeline. That trade-off can still be worth accepting, since the same record also spared the judges false alarms on decoys.

How the record is presented can matter, and not in the same way for every judge. For three of our judges a true warning that the record was incomplete restored most catches on its gaps, but it also brought back the false alarms on decoys that the record had removed, and the fourth judge recovered few. A builder considering such a warning should test it on their own judge and items, count false alarms as well as catches, and make sure the warning is true.

Scoring should use reasons. Any-finding scoring inflates false alarms on correct answers and cannot distinguish a judge that raises an irrelevant objection from one that says nothing. Where the record lists both replacements and non-replacing updates, as the RFC index does, date controls and decoys come at no extra cost.

If the replacements in a target collection tend to look unlike the documents they replace, catches on such pairs should be measured separately, since our judges were weakest there.

8 Limitations and threats to validity

Two draws of dropped links, few items.

Each draw gives each judge 24 gap faults; the two draws share 12, so the gap result rests on 36 distinct links. The drop held on both draws for every judge, but the pilot showed how much a judge's result can depend on which links are dropped (§5.2), and differences between judges of a few items remain within

what such variation could produce. The presentation conditions were run on the first draw only, and Claude's presentation effects are not significant.

Reused replies, single runs and decoding.

Text-only replies on the first 24 links were reused from the pilot for all four judges, and the second draw reuses the first draw's text-only replies, so H2a is partly a re-analysis of pilot replies; on the gap faults where every reply is new, the drop is of similar size (§5.2). Hosted judges ran once each, through a proxy that sets and does not report temperature and seed; in the pilot, a repeated block changed 30 of 72 replies but only one catch. The reused replies came from an earlier form of the proxy call, and Gemma's client sometimes retried a cut-off reply without the API's JSON mode (§4).

The decoys are a partial control.

They are matched on dates, title similarity and, for most pairs, IETF area, but not on abstract overlap, which differs sharply between replacements and updates. Some decoys are new protocol versions, for which flagging the older RFC is a defensible reading of the question even though the publisher's record says it was not replaced. The similarity analysis is exploratory, post hoc and small.

Construct.

The evidence is titles, dates and abstracts, not full documents. The error tested is always the citation of a whole replaced document; partial amendments and chains are not covered. Links are dropped at random, while real records fail in patterns (recent amendments, particular issuers). The question names the old document's title, which may provoke some of the non-currency findings on controls.

Generalisation.

One corpus of technical standards, chosen because its publisher maintains an exact record, and four judges. Legal collections add problems this test does not address, as the case study illustrates: partial amendments, records extracted by machine, and answer keys that people disagree about. The title of the paper asks a general question; the answer we give is limited to these judges, prompts and documents.

Mechanism.

The presentation conditions keep the record fixed and vary single features, one prompt variant each. They can show that presentation changed what a judge did; they cannot show why.

Provenance.

The harness and the text were produced with an AI assistant from the same developer as one of the judges, Claude Opus 4.5 (§4). All judges received identical prompts and items, and the gold comes from the publisher. The plan was written by the same team after the pilot and recorded internally rather than registered publicly.

9 Conclusion

We asked what LLM judges rely on when they check whether a cited document is still in force. On a corpus with an exact publisher record, the judges we tested matched a lookup when the record was correct, and on links we removed from the record they caught markedly fewer replaced citations than they did from the texts alone, usually without comment, on both of two random choices of removed links. This happened even when the record line said, truthfully, only that no replacement was recorded. Labelling the record as unverified or incomplete brought three of the four judges most of the way back to the texts, their false alarms included, and the fourth part of the way. An evaluation with a correct record and any-finding scoring would have shown none of these behaviours. The test needs only a table of replacements, and we release it so that others can run it on their own judges and collections.

10 Data and code availability

The repository (https://github.com/unnatibshah/what-do-llm-judges-verify) holds the harness (`evalkit.py`), the run and analysis scripts, the sampled links and decoys, every judge reply with its block metadata (model version, context, code hash), and the scripts that regenerate every number, table and

Table A1 Where the record is silent. Left: right-reason catches on all faults with no record (`both-texts`), a correct record line (`status`) and a record with half its links dropped (`status-damaged`, seed 0); a lookup in the damaged record catches exactly half. Right: the faults whose link was dropped, with no record, with the damaged record (the line reads "in force"), and with the line worded "no replacement recorded"; same items in every column. Bare: damaged-record replies on these faults that pass the answer with no finding. Qwen 32B ran 48 links, the others 24.

	All faults			Faults whose link was dropped			
Reviewer	No record	Correct	Damaged	No record	"In force"	"No replacement"	Bare
Qwen 32B	42/48	48/48	33/48	22/24	9/24	5/24	6
Gemma 4 26B	16/24	24/24	12/24	10/12	0/12	0/12	12
GPT-OSS 120B	16/24	18/24	15/24	8/12	6/12	–	4
Gemini 3.1 Pro	22/24	24/24	12/24	12/12	0/12	3/12	12
GPT-5.5	23/24	–	16/24	11/12	4/12	6/12	8
Claude Opus 4.5	19/24	24/24	22/24	11/12	10/12	9/12	1

Table A2 Currency discrimination by context (points): right-reason catch on faults minus validity false alarms on date controls, which cite the same old document as of a date before it was replaced. Mean over links.

Context	Qwen 32B	Gemma 4 26B	GPT-OSS 120B	Gemini 3.1 Pro	GPT-5.5	Claude Opus 4.5
`cite-only`	−38	+12	–	+46	–	+21
`cited-text`	−69	–	–	–	–	+25
`both-texts`	−2	+67	+67	+92	+96	+71
`both-masked`	−19	+62	+54	+83	+92	+75
`both-explicit`	−2	–	+88	+100	+100	+92
`status`	+77	+100	+71	+100	–	+100
`status-damaged`	+46	+50	+62	+50	+67	+92
`norecord-damaged`	+27	+50	–	+62	+75	+83
`check-first`	+23	+75	–	+100	–	+88
`few-shot`	+79	–	–	–	–	+75

figure in this paper from those replies: `analyse_final_e123.py` rescores every reply from its raw text, checks the stored scores and regenerates §5 and Appendix B; `python -m suryabench.evalkit` does the same for the pilot. The RFC Editor index is cited with its retrieval date and content hash (RFC Editor 2026) and included unchanged. The hypotheses, tests and readings were committed to the project's development repository before the runs, and the amendment adding the second draw before any call on it. That repository is private; the release contains the plan with its amendments and a provenance file listing the commit and time at which the plan, each amendment and each block of results were recorded. For the case study, the regulatory documents are released as source URLs with content hashes, and the cases, the fixed answers, the record before and after adjudication and the annotation sheets are released with the code. Code is under the Apache-2.0 licence; the judge replies, the analysis outputs and the case study's cases and labels are under CC BY 4.0.

Appendix A The pilot

Six judges (Qwen 2.5 32B, Gemma 4 26B, GPT-OSS 120B, Gemini 3.1 Pro, GPT-5.5 and Claude Opus 4.5) ran up to ten contexts on the first 24 links before the final design was fixed; Qwen 2.5 32B ran all 48. Table A1 gives the pilot's gap result, on the 12 gap faults per judge produced by a damaged record over 24 links, and Table A2 the currency discrimination against date controls in every context. The generated pilot tables use the pilot's own vocabulary ("reviewer" for judge, "deference" for the gap contrast). The pilot's gap faults overlap only partly with the final run's (§5.2).

Runs.

Hosted models ran once through the Kaggle Benchmarks proxy, which sets temperature and seed itself; GPT-5.5's two damaged-record blocks ran after a change to the proxy call (an output cap, and the system and user text sent as two messages), with the prompt text unchanged. An accidental repeat of one block (Gemini 3.1 Pro, citation only; the first run is of record) differed by one catch. Every block records the model version, the context and the harness's code hash. A reply that cannot be parsed counts as a miss on a fault and a false alarm on a control; GPT-OSS had two to four such replies per block, the other reported judges at most two. Three further runs stopped before the deference pair of contexts and are not analysed; their blocks are in the repository.

Table A3 Discrimination against current-document controls (points): right-reason catch minus validity false alarms on answers citing the replacing document. Mean over links.

Context	Qwen 32B	Gemma 4 26B	GPT-OSS 120B	Gemini 3.1 Pro	GPT-5.5	Claude Opus 4.5
`cite-only`	−31	+12	–	+46	–	+21
`cited-text`	−69	–	–	–	–	+21
`both-texts`	+40	+67	+62	+92	+96	+75
`both-masked`	+35	+62	+50	+83	+92	+75
`both-explicit`	+46	–	+83	+100	+100	+100
`status`	+100	+100	+75	+100	–	+100
`status-damaged`	+69	+50	+62	+50	+67	+92
`norecord-damaged`	+60	+50	–	+62	+75	+88
`check-first`	+75	+75	–	+100	–	+88
`few-shot`	+88	–	–	–	–	+75

Items.

Two of the 48 replacing RFCs name the old RFC in their title (RFC 7142, RFC 6082); titles are not filtered, so these two links are explicit for Qwen 32B, the one judge that ran all 48. The sentence filter of `both-texts` also removes some sentences that mention updates in general.

Links without a shared title.

On the 6 of the first 24 links whose successor has a different title, with both texts, GPT-5.5 caught 6/6, Gemini 5/6, Claude 3/6, GPT-OSS 2/6 and Gemma 1/6. Masking the document numbers changed little for the frontier judges (Gemini 20/24, Claude 20/24, GPT-5.5 22/24) and cost GPT-OSS a few catches (13/24).

Bare passes.

A reply on a fault is a bare pass when it has no finding at all; a reply with findings is coded by whether its rationale mentions the record (a keyword rule, post hoc).

Controls citing the successor.

Table A3 gives discrimination against controls that cite the replacing document. Given the documents, the frontier judges, GPT-OSS and Gemma rarely gave a validity reason on these controls; Qwen 32B gave one on 23/48. Other findings on them ("not applicable", "unsupported") are expected, because the question names the old document's title.

Appendix B Gap faults item by item

Table B4 gives, for each judge, the outcome on every gap fault under no record and the four presentations of the damaged record. Every reply, with its raw text, is in the repository (`50_analysis/results/evalkit_rfc_v2/`; second draw in `evalkit_rfc_v2_seed1/`). The outcome of every link under both draws, with the draws it belongs to, is in `evalkit_rfc_v2/analysis/seed1_links.csv`.

Appendix C Prompt

Figure C1 gives the system text and the full prompt for one gap fault under the authoritative line, as sent, and the record block of each damaged context. The other contexts differ only in the record block, which is absent from `both-texts`.

Table B4 Per-item outcomes on the gap faults (✓ caught for the right reason, · missed, – not run). Within each judge, the five marks are: no record, "in force", "no replacement recorded", unverified note, record with warning.

Link (RFC)	Gemma 4 26B	GPT-5.5	Gemini 3.1 Pro	Claude Opus 4.5
6312→6342	· · · · ·	· · · ✓ ✓	✓ · · · ✓	· · · · ·
8773→9973	· · · · ·	✓ · · ✓ ✓	· · · · ·	✓ · ✓ ✓ ✓
5046→7145	✓ · · · ·	✓ · · ✓ ✓	✓ · · ✓ ✓	✓ ✓ ✓ ✓ ✓
8407→9907	✓ · · ✓ ✓	✓ · · ✓ ✓	✓ · · ✓ ✓	✓ ✓ ✓ ✓ ✓
3205→9205	· · · · ·	✓ · · ✓ ✓	✓ · · ✓ ✓	✓ · · · ·
2821→5321	✓ · · · ·	✓ · ✓ ✓ ✓	✓ · · ✓ ✓	✓ ✓ · ✓ ✓
4402→7802	✓ · · ✓ ·	✓ ✓ ✓ ✓ ✓	✓ · ✓ ✓ ✓	✓ · ✓ ✓ ✓
5157→7707	· · · · ·	✓ · · ✓ ✓	✓ · · · ✓	· · · · ·
2837→4044	· · · · ·	✓ · · ✓ ✓	✓ · · · ✓	· · · · ·
2621→4671	· · · · ·	✓ · · ✓ ✓	✓ · · · ✓	· · · · ·
2858→4760	✓ · · · ✓	✓ ✓ ✓ ✓ ✓	✓ · · ✓ ✓	✓ ✓ ✓ ✓ ✓
2478→4178	✓ · · · ·	✓ ✓ ✓ ✓ ✓	✓ · · ✓ ✓	✓ · · ✓ ✓
1657→4273	· · · · ·	✓ · · ✓ ✓	✓ · · ✓ ✓	✓ · · · ✓
1142→7142	· · · · ·	· · · ✓ ·	✓ · · ✓ ✓	✓ · ✓ ✓ ✓
8941→9651	✓ · · · ✓	✓ ✓ ✓ ✓ ✓	✓ · ✓ ✓ ✓	✓ ✓ · ✓ ✓
5430→6460	✓ · · · ·	✓ · · ✓ ✓	✓ · ✓ ✓ ✓	✓ ✓ ✓ ✓ ✓
2482→6082	· · · · ·	✓ · · ✓ ✓	✓ · · ✓ ✓	✓ ✓ ✓ ✓ ✓
3984→6184	✓ · · · ·	✓ ✓ ✓ ✓ ✓	✓ ✓ ✓ ✓ ✓	✓ ✓ ✓ ✓ ✓
4244→7044	✓ · · · ·	✓ ✓ ✓ ✓ ✓	✓ · · ✓ ✓	✓ ✓ ✓ ✓ ✓
3230→9530	· · · · ·	✓ · · ✓ ✓	✓ · · ✓ ✓	· · · ✓ ✓
7613→8265	✓ · · · ✓	✓ · ✓ ✓ ✓	✓ · · ✓ ✓	✓ ✓ ✓ ✓ ✓
2767→6535	· · · · ·	✓ · · ✓ ✓	· · · · ·	· · · · ·
8843→9143	✓ · · · ✓	✓ · ✓ ✓ ✓	✓ · · ✓ ✓	✓ · · ✓ ✓
2910→8010	✓ · · · ·	✓ ✓ ✓ ✓ ✓	✓ · · ✓ ✓	✓ · · ✓ ·

Appendix D Case study: Indian rooftop-solar regulation

Status: descriptive only. The test was first developed on this setting, where no curated consolidation exists and the amendment record had to be extracted from the documents by an AI pipeline. Its answer key failed the quality checks planned for it: a person applying the right-reason rule agreed with the automatic scorer on 70% of dev replies, against a planned threshold of 95%, and when people read whole case files, many of the fixed "correct" answers turned out to be incomplete. None of its numbers is used as evidence for the hypotheses or findings of §5. It is reported because its failures, annotation disagreements and negative results show what the test needs from an uncurated amendment record; its planned analyses and readings are kept as they were fixed, so that nothing is selected after the fact. The full report (reference data and their audit, conditions, planned analyses and results, readings fixed in advance, prompts, per-clause results, annotation agreement, deviations from the plan and the data statement) is in the supplementary material.

Two consequences follow for reading the numbers below. False alarms on correct answers cannot be taken as the judge's errors. For example, Claude Opus 4.5 raised some finding on 37/42 correct answers when given the status line, but most of its findings were not about validity (44/55), and only a corrected key could say whether they were wrong. Likewise, the large differences between any-finding and right-reason catches in this study are real outputs on this key, but we do not use them to support §5.4, which rests on the RFC data. The text below is carried over from the study's own report, written before the RFC study, and uses its terms (reviewer for judge, registry for record, clause for document).

D.1 Setting

Documents and record.

The corpus holds 111 regulatory instruments on rooftop solar from 8 Indian jurisdictions, superseded ones included. No consolidated record exists, so an AI pipeline extracted one from the documents: 186 clause-level links, each from an old clause to the clause that changes it, with a relation and an effective date. Two authors checked a random sample of the links blind; 5 of 34 were judged to be no change at

Table D5 The findings across all models, on the same items (pooled k/n over S-both faults or matched correct answers; 50%-dropped arms pooled over each model's damage seeds). Models are ordered by size; the last columns were added after the analysis plan was fixed. Gains and losses compare each fault and seed with the damaged lookup. The dropped-link catch is clause-averaged over the faults the intact registry catches and the damaged one drops.

Finding	Llama 8B	Qwen 7B	Qwen 14B	Gemma 4 26B	Flash-Lite	Qwen 32B	Claude Opus 4.5
Right-reason catch, both versions (R2)	0/42	5/42	13/42	9/42	19/42	4/42	34/42
Any-finding catch, R2	24/42	14/42	26/42	32/42	35/42	35/42	42/42
Scoring gap at R2, any minus right reason (points)	+57	+21	+31	+55	+38	+74	+19
Right-reason catch, status line (R4)	3/42	2/42	32/42	25/42	33/42	29/42	35/42
Correct answers flagged, R4	19/42	2/42	10/42	20/42	19/42	23/42	37/42
R4 findings on correct answers not about validity	13/25	0/2	4/11	28/30	19/22	21/24	44/55
Catches gained / lost vs lookup, 50% dropped	+0 / −15	+2 / −15	+13 / −13	+25 / −24	–	+4 / −8	+14 / −0
Catch on dropped links, R2 → R4-damage, % (clauses)	0 → 0 (6)	33 → 17 (6)	41 → 7 (9)	32 → 31 (9)	–	13 → 0 (9)	92 → 67 (6)

all, and every link the test relies on was checked and corrected (supplementary material, §S1.1). The record therefore asserts few wrong links, but nothing measures how many it misses, as in practice.

Items.

A case is a synthetic solar application (a bill, an installer proposal and a utility record) with a date and a jurisdiction, labelled by rule functions whose parameters are quoted from the clauses. Its answer is an approve-or-reject decision with the clauses it rests on. A fault cites a replaced clause in place of its successor; its matched control is the correct decision citing the current clause. The case study has no date control, because each case has one date. People audited the answer key before the test set was sealed: the AI-built key reached a different outcome from the adjudicated one on 18 of 60 candidate cases, which were dropped. The sealed set has 42 cases on 14 distinct replaced clauses from 6 jurisdictions; the record links 11 of the 14, so the other replaced clauses are gaps the record already has.

What the reviewer sees.

The main conditions are three contexts of the framework: both versions of the clause (R2, `both-texts`), both versions plus the record's status line for each cited clause (R4, `status`), and the same line from a damaged record (R4-damage, `status-damaged`; R4n-damage, `norecord-damaged`). The record is damaged by dropping 25% or 50% of its links, or adding 10% spurious ones, over 5 seeds. A deterministic lookup in the same record (R5, no language model) flags exactly the clauses the record marks as replaced; it is the reference for what the record alone achieves. Further conditions (the old clause alone, source metadata, the amending instrument's words) are in the supplementary material (§S1.2).

Reviewers and scoring.

Seven reviewers judged the same items: `llama3.1:8b`, `qwen2.5:7b`, `qwen2.5:14b`, `gemma-4-26b-a4b-it`, `gemini-3.5-flash-lite` (on R0, R2 and R4 only), `qwen2.5:32b` and Claude Opus 4.5, with greedy decoding. A fault is caught when a finding names the replaced clause with the reason "not in force". The unit of inference is the distinct replaced clause: rates are averaged within a clause, then over clauses. The questions, analyses and readings were fixed in a dated, version-controlled plan before any model saw a test item; the planned tests, all reported, are in the supplementary material (§S1.3 and S1.4).

D.2 Descriptive results

Table D5 summarises the findings across the seven judges on the same items. Figure D2 compares the judge given the record's status line with a lookup in the same record as the record degrades.

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```
You review answers that cite documents. Be precise and brief. Respond
  in strict JSON.

[QUESTION]
As of 2026-10-01, which document specifies "Mobile Networks
  Considerations for IPv6 Deployment"?
[ANSWER UNDER REVIEW]
The document that specifies this is RFC6312, "Mobile Networks
  Considerations for IPv6 Deployment" (2011-07).
[EVIDENCE]
[RFC6312] Mobile Networks Considerations for IPv6 Deployment
  (published 2011-07)
Mobile Internet access from smartphones and other mobile devices is
  accelerating the exhaustion of IPv4 addresses. IPv6 is widely seen
  as crucial for the continued operation and growth of the Internet,
  and in particular, it is critical in mobile networks. This document
  discusses the issues that arise when deploying IPv6 in mobile
  networks. Hence, this document can be a useful reference for service
  providers and network designers. This document is not an Internet
  Standards Track specification; it is published for informational
  purposes.

[RFC6342] Mobile Networks Considerations for IPv6 Deployment
  (published 2011-08)
Mobile Internet access from smartphones and other mobile devices is
  accelerating the exhaustion of IPv4 addresses. IPv6 is widely seen
  as crucial for the continued operation and growth of the Internet,
  and in particular, it is critical in mobile networks. This document
  discusses the issues that arise when deploying IPv6 in mobile
  networks. Hence, this document can be a useful reference for service
  providers and network designers. This document is not an Internet
  Standards Track specification; it is published for informational
  purposes.
[AMENDMENT RECORD LOOKUP OF THE CITED DOCUMENT]
- RFC6312: status on 2026-10-01: in_force
[TASK]
Decide whether the answer is valid as of the date in the question. For
  each problem give a finding naming the cited document and one reason
  from:
- superseded: the cited document had been replaced or obsoleted by a
  later one on the date asked about
- not_yet_published: the cited document did not yet exist on the date
  asked about
- not_applicable: the cited document does not address the question
- content_error: the answer misstates what the document says
- unsupported: the evidence does not support the answer
- other: any other problem
Reply as JSON: {"verdict": "PASS" or "FLAG", "findings": [{"doc_id":
  "...", "reason": "...", "rationale": "..."}]}
```

```
[AMENDMENT RECORD LOOKUP OF THE CITED DOCUMENT]
- RFC6312: status on 2026-10-01: in_force

[AMENDMENT RECORD LOOKUP OF THE CITED DOCUMENT]
- RFC6312: status on 2026-10-01: no_replacement_recorded

[UNVERIFIED NOTE FROM AN AUTOMATED TAGGER ABOUT THE CITED DOCUMENT]
- RFC6312: status on 2026-10-01: in_force

[AMENDMENT RECORD LOOKUP OF THE CITED DOCUMENT (this record is
  incomplete: it omits some replacements, so a document it lists as in
  force may have been replaced)]
- RFC6312: status on 2026-10-01: in_force
```

Fig. C1 Top: system text and prompt for one gap fault under the authoritative line. Bottom: the record block of each damaged context, in table order.

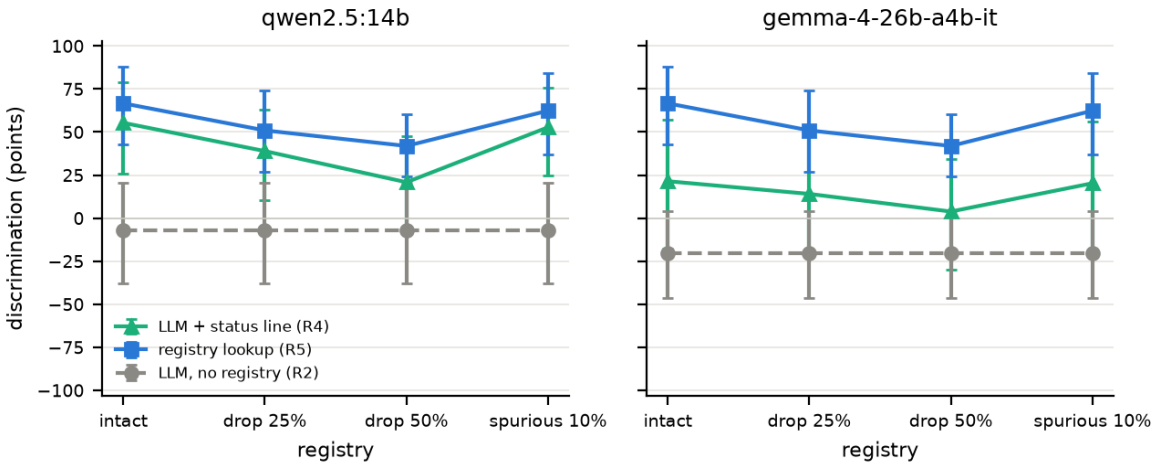


Fig. D2 Case study: discrimination (right-reason catch minus false alarms on matched correct answers) as the record degrades, for the judge given the status line (R4), a lookup in the same record (R5), and the judge with no record (R2). Clause-averaged, damage seeds averaged within clause, 95% bootstrap intervals over 14 clauses.